

Compiler Design

CO-2 Assessment Tool-3

S.A. Mohammed Fareeth
192511288

i) a) Trading System Expression Parsing

$$E \rightarrow E + E \mid E - E \mid E * E \mid id$$

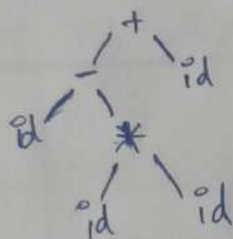
Input: $id - id * id + id$

(i) Identity Ambiguity:

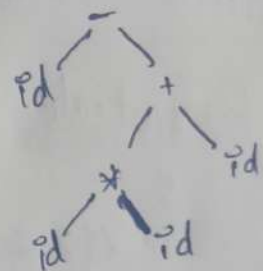
Grammar is ambiguous because the expression can be parsed in more than one way

(ii) Two Parse Trees:

Tree 1: $((id - (id * id)) + id)$



Tree 2: $(id - ((id * id) + id))$



(iii) Redesign Grammar:

$$E \rightarrow E + T \mid E - T \mid T$$

$$T \rightarrow T * F \mid F$$

$$F \rightarrow id$$

(iv) Correct Parsing

$$id - id * id + id$$

$$\Rightarrow id - (id * id) + id$$

$$\Rightarrow (id - (id * id)) + id$$

(v) Performance Considerations

- * Use an unambiguous grammar
- * Apply operator precedence
- * Use an efficient parser (LALR (1)/LR)
- * Reduces parsing time and errors

b) IoT Gateway Nested Sensor Packets

$S \rightarrow \langle S \rangle S \mid \text{data}$

Input: $\langle \langle \text{data} \rangle \text{data} \rangle$

(i) Explain Nested Parsing

This parser matches every $\langle$ with a corresponding $\rangle$ and supports nested packets

(ii) Demonstrate Parsing:

S
 $\Rightarrow \langle S \rangle S$
 $\Rightarrow \langle \langle S \rangle S \rangle S$
 $\Rightarrow \langle \langle \text{data} \rangle \text{data} \rangle$

(iii) Identify Incomplete Packet Issues

- * Missing $\rangle$
- * Missing $\langle$
- * Unexpected end of input
- * Invalid nesting

Suggest Streaming Parser Design

- Use a stack to track <
- Process input token by token
- Wait for remaining data if packet is incomplete

(v) Explain Error Detection

- Report unmatched < or >
- Detect Invalid nesting immediately
- Reject malformed packets

(vi) Nested Conditionals (Dangling Else)

$S \rightarrow \text{if } E \text{ then } S \text{ else } S \mid \text{if } E \text{ then } S \mid \text{id}$

$E \rightarrow \text{id}$

Input: if id then if id then id else id

(vii) Identify Dangling Else Problem:

The else can match either the inner or outer if, making the grammar ambiguous

(viii) Rewrite Grammar

$S \rightarrow M \mid U$

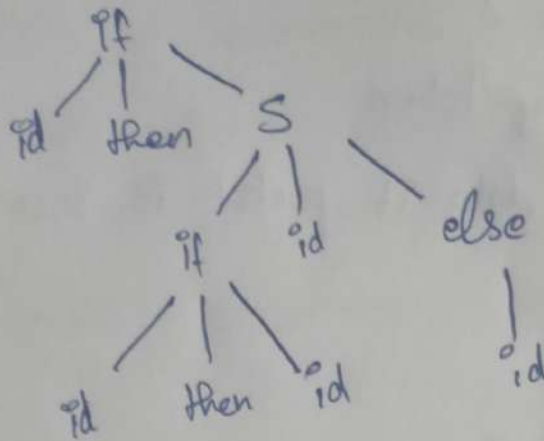
$M \rightarrow \text{if } E \text{ then } M \text{ else } M \mid \text{id}$

$U \rightarrow \text{if } E \text{ then } S \mid \text{if } E \text{ then } M \text{ else } U$

(ix) Demonstrate Parsing

The else is matched with the nearest unmatched if
if "if id then (if id then id else id"

Construct Parse tree



Explain Compiler Behavior

- * Most compiler associates else with the nearest unmatched if
- * This removes ambiguity and ensures consistent parsing